

ANDREA ZIGNOLI

M.Eng, PhD | Applied Research Scientist & Consultant in Sport Technology

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 Andrea-Zignoli



PRESENTATION

I'm an applied research scientist with 10+ years of experience developing AI-driven solutions for sports performance analysis, combining academic rigor with real-world impact.

CURRENT POSITIONS

Applied Research Scientist - Product Expert

Athletica Inc.

 Apr 2020 – Ongoing  Remote location

- Part-time back-end lead and product expert.
- Responsible for the development of the data processing & analysis for a web-based endurance training platform.

Consultant

Tymewear

 November 2024 – Ongoing  Remote location

- AI features development - ML models - back-end development.

Consultant

Enhance-d

 June 2024 – Ongoing  Remote location

- Development of DL models from sport-wearable data sources.

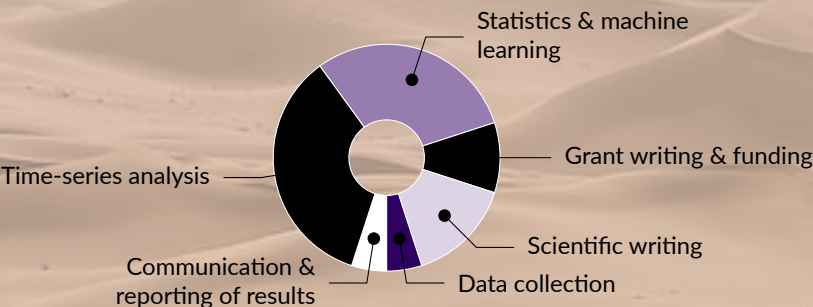
STACK I USE

 Machine Learning Development
Python: Scikit Learn - Pandas - Keras - Tensorflow

 Containerization & Backend Development
Docker | Python: Flask | Shiny (R)

 Cloud Deployment
Heroku | AWS Lightsails

HOW I SPEND MY TIME



EDUCATION

PhD in Science of Physical Exercise and Human Movement

University of Verona, Verona, Italy

 Jan 2013 – May 2016

M.Eng. in Mechatronics

University of Trento, Trento, Italy

 Sept 2008 – March 2011

METRICS (SCOPUS)

 43 peer-reviewed publications
41% first-author | 8% single-author

 329 Citations (H-index 10)
20.5% (8 documents) in the top 25% most cited documents worldwide

CODING SKILLS

Python R Matlab

STATISTICS & ML

(RM)-ANOVA Multi-linear models

PCA SVM

Deep learning

CNN LSTM GAN

DATA PROCESSING

GPS Power meters EMG NIRS

Lactate monitors Pressure insoles

CPET IMU Motion capture

Force platforms Optitrack HRV

LANGUAGES

Italian
English



WORKING EXPERIENCE

Teaching Fellow

Department of Cellular, Computational and Integrative Biology - CIBIO, University of Trento

📅 Dec 2021 – Ongoing 📍 Trento, I

- Instructor for the 3-credit module on Modeling within the Technology and Innovation in Sports course in the Master's program in Exercise Science and Human Movement

Fellow Researcher

Department of Industrial Engineering, University of Trento

📅 Dec 2020 – Ongoing 📍 Trento, I

- Development and deployment of mathematical models for sport technology applications.

Associate Editor

Sports Engineering, the official journal of the International Sports Engineering Association (ISEA)

📅 Oct 2021 – Ongoing 📍 Remote location

- Handling and reviewing research papers.
- Creating partnerships with sport tech companies, governing bodies, and organising committees.

Applied Research Scientist

Supersapiens

📅 Apr 2020 – Apr 2024 📍 Remote location

- Data analysis and report creation (internal, coaches, athletes, users)
- Product design and scientific support

Functional evaluation centre

CeRiSM Sport Mountain and Health Research Centre

📅 Apr 2012 – Oct 2017 📍 Trento, I

- Data collection and athlete functional evaluation
- Scientific writing

KEY PUBLICATIONS

📄 Journal Articles (selection)

- Zignoli, Andrea, Federico Y Fontana, et al. (2023). "Association between pre-exercise food ingestion timing and reactive hypoglycemia: Insights from a large database of continuous glucose monitoring data". In: *European journal of sport science* 23.12, pp. 2340–2348.
- Zignoli, Andrea, Antoine Godin, and Laurent Mouroto (2023). "Indoor running temporal variability for different running speeds, treadmill inclinations, and three different estimation strategies". In: *PloS one* 18.7, e0287978.
- Zignoli, Andrea, Alessandro Fornasiero, et al. (2020). "Oxynet: a collective intelligence that detects ventilatory thresholds in cardiopulmonary exercise tests". In: *European Journal of Sport Science*, pp. 1–38.

SIDE PROJECTS I LEAD



Oxynet
www.oxynet.net

Process and interpret cardiopulmonary exercise tests with AI technologies.



Workout Reserve
github.io

The Athletica Workout Reserve Connect IQ App for endurance athletes.



CURVA
bitbucket.org

Cycling trajectories optimized using optimal control.



Drone Footage
github.com

Processing and analysis of drone footage for cycling applications.



PRISV
bitbucket.org

Running step frequency analysis using IMUs and deep learning models.

REFEREES

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Revelstoke, B.C., Canada

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